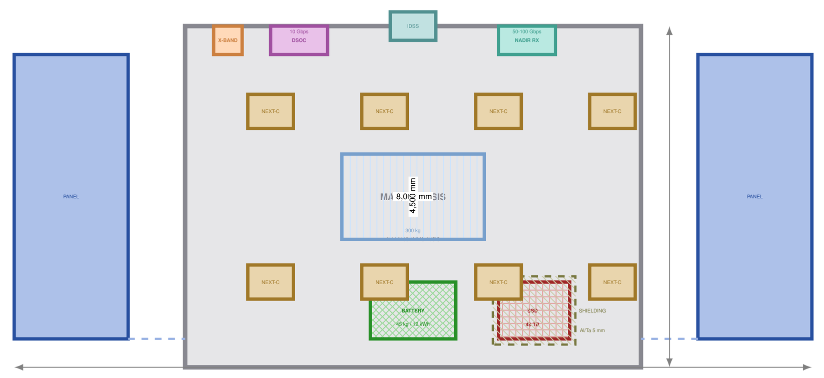
TUG-BIG: ION TRACTOR — VLBI RELAY NODE

8x NEXT-C | SEP | SSD 48 TB | DSOC + NADIR RX + RF

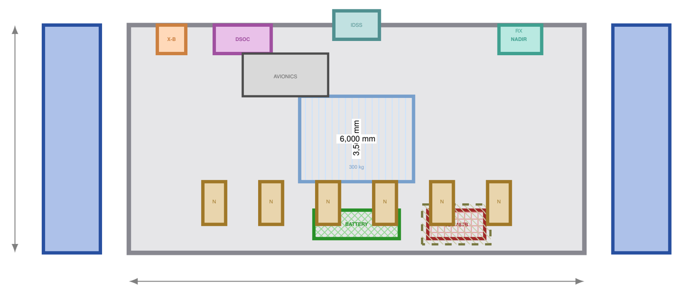
DRY MASS: 1,610.5 kg | XENON: 300 kg | TOTAL: 1,910.5 kg

ARCHITECTURE VLBI → TUG RELAY → EARTH

SIMULATION | CDR READY



PLAN VIEW (TOP)



PROFILE VIEW (SIDE)

MISSION:	L-RAIL (NASA/COPAG/PhysPAG)	REV: D
DRAWING:	DWG-TUGBIG-00 | SHEET 1 of 1	DATE: 2026-08-06
TITLE:	TUG-BIG: ION TRACTOR x8 (VLBI RELAY)	SHEET 1/1
SCALE: 1:40	UNITS: mm / kg	FORMAT: A1 L

L-RAIL

ID	Link	Direction	Rate	Band	Role
TUG-640	Surface → TUG	RX (nadir)	50–100 Gbps	Optical 1550 nm	Primary: receive VLBI science
TUG-630	TUG → Earth	TX (Earth-pointing)	10 Gbps	Optical 1.55 μm	Primary: downlink to Earth
TUG-650	TUG → Earth	TX (Earth-pointing)	500 Mbps	X-band 8 GHz	Backup if DSOC fails
TUG-650b	Earth → TUG	RX	10 Mbps	S-band 2 GHz	TT&C + commands

Justification for (X-band): The optical DSOC requires pointing $<5\ \mu$rad. In case of optical gimbal failure, laser degradation, or temporary attitude loss, the X-band (500 Mbps) allows:
- Downloading critical scientific data at reduced rate (500 Mbps vs 10 Gbps).
- Maintaining telemetry and commands (TT&C) in S-band continuously.
- Does not depend on fine pointing: omnidirectional or low-gain patch antenna.

Note: The X-band does NOT replace the DSOC. It is only for emergency. Full download time at 500 Mbps: 48 TB $\times$ 8 / 0.5 Gbps $\approx$ 768,000 s $\approx$ 213 h. Only viable for selective critical data.

Phase	Action	Xenon (kg)	Duration
1	HEO loading: coupling with payload (ASL-H, VADS, cartridges)	—	$\sim$2 days
2	Translunar injection: 8 NEXT-C engines at 100%	$\sim$270	$\sim$14 days
3	Arrival at lunar polar orbit ($\geq$800 km)	—	—
4	Payload release: IDSS releases cargo to ASL	—	$\sim$1 day
5	Return to HEO: 1-2 engines, low power	$\sim$9	$\sim$10 days
6	Relay mode (post-deployment): lunar polar orbit	0 (solar)	Indefinite

Relay mode (post-deployment): Once the 5 transport trips are completed, the TUG-BIG remains in lunar polar orbit as a relay node. It receives VLBI data from the surface during passes of $\sim$30 min every $\sim$95 min and downloads them to Earth via DSOC. Two TUG-BIG in polar orbit take turns (*relay*) for continuous coverage.
DTN Protocol: Custody transfer (CCSDS 734.2-B-1). The TUG confirms reception of each bundle. If no ack, the surface retains and retransmits on the next pass.
Dual buffer: Surface SSD (32 TB, DWG-BACK-03) + TUG SSD (48 TB). Combined autonomy of $\sim$6.9 h at 25.6 Gbps without downlink to Earth.

Trip	Cargo	HEO Mass (kg)	Xe Outbound (kg)	Xe Return (kg)	Xe Total (kg)	Margin (kg)
1	ASL-H + 1 Cartridge	13,165	270.3	8.9	279.2	+20.8
2	VADS + ASL-Std	6,253	130.8	8.9	139.7	+160.3
3	3 Cartridges (2, 3 and 4)	12,015	246.1	8.9	255.0	+45.0
4	MrN + 1 Cartridge	7,765	156.5	8.9	165.4	+134.6
5	3 Cartridges (6, 7 and 8)	12,015	246.1	8.9	255.0	+45.0
					TOTAL XENON CONSUMED (5 trips)	**1,094.3 kg**
					TOTAL XENON AVAILABLE (5 trips $\times$ 300 kg)	**1,500.0 kg**
					GLOBAL MARGIN	**+405.7 kg**

Impact of Rev. D on dry mass: +15.0 kg (from 1,595.5 to 1,610.5 kg). Variation $<1\%$. Does not affect the xenon budget or payload capacity (11,500 kg).
The simulated orbital performance in TUG-BIG-v1 remains unchanged because Δm = 15 kg is negligible compared to 11,500 kg of payload.

ID	Subsystem	Mass (kg)	Material	Standard
TUG-100	Dry structure	500	Al-Li 2195 / MMC Al/SiC	NASA-STD-6016
TUG-200	Xenon tank	150	Ti-6Al-4V / CFRP	NASA-STD-6016
TUG-300	Xenon (propellant)	300	Xe (high pressure)	NASA-STD-3001
TUG-400	NEXT-C engines (x8)	480	Molybdenum (Mo) grids	NASA-STD-5017
TUG-500	Solar panels (x2)	200	Triple-junction GaAs	NASA-STD-5003
TUG-600	Avionics + GNC + Comms	80	RAD750, SpaceWire	IEEE 1355
TUG-610	**SSD 48 TB rad-hard (UPGRADED)**	**1.0**	**NAND Flash + ECC**	**NASA-STD-8739.8**
TUG-620	**Radiation shielding (Al/Ta, expanded)**	**2.5**	**Aluminum + Tantalum**	**NASA-STD-5003**
TUG-630	**DSOC optical link (Earth TX)**	**5.0**	**1.55 μm, 10 Gbps**	**CCSDS 734.2-B-1**
TUG-640	**NADIR optical terminal RX (surface)**	**10.0**	**1550 nm, 50–100 Gbps, 300 mm**	**CCSDS 734.2-B-1**
TUG-650	**RF X-band + S-band TT&C**	**4.0**	**GaN PA + patch ant + diplexer**	**CCSDS 401.0-B**
TUG-700	IDSS (docking)	50	Stainless + SmCo5	AIAA S-114
TUG-800	Thermal control	85	Loop heat pipes + radiators	NASA-STD-5003
TUG-900	Li-ion battery 12 kWh	45	Li-ion NMC	NASA-STD-3001
	TOTAL DRY MASS (REV. D)	**1,610.5 kg**		
	TOTAL MASS WITH XENON	**1,910.5 kg**		

TUG-640: NADIR RX

Parameter	Value
Function	RX from surface
Wavelength	1550 nm
RX rate	50–100 Gbps
Aperture	300 mm
Acquisition FOV	± 60 nadir
Pointing	2-axis gimbal
Precision	$<5\ \mu$rad
Protocol	DTN Bundle
Mass	10.0 kg

TUG-630: DSOC TX

Parameter	Value
Function	TX to Earth
Wavelength	1.55 μm
Downlink rate	10 Gbps
Uplink rate	1 Gbps
Optical power	5 W
Aperture	200 mm
Modulation	QPSK
Protocol	CCSDS 734.2-B-1
Mass	5.0 kg

TUG-650: RF

Parameter	Value
Function	Earth backup
Band	X-band 8 GHz
TX rate	500 Mbps
TX power	5 W
Antenna	Patch / 0.3 m
TT&C	S-band 2 GHz
TT&C rate	10 Mbps
Modulation	QPSK + LDPC
Mass	4.0 kg

ID	Connection	Type	Standard
1	Solar panels → Battery	Power (100 V DC)	NASA-STD-5003
2	Battery → Avionics	Power (28 V DC)	NASA-STD-3001
3	Avionics → SSD	Data (SpaceWire)	IEEE 1355
4	Avionics → NEXT-C	Control (MIL-STD-1553B)	MIL-STD-1553B
5	Avionics → IDSS	Data/Power	NASA-STD-7001
6	Avionics → DSOC (TUG-630)	Data (SpaceWire)	IEEE 1355
7	DSOC → Optical TX antenna	Optical fiber	CCSDS 734.2-B-1
8	NADIR RX (TUG-640) → SSD	Data (high rate)	CCSDS 734.2-B-1
9	Avionics → X-band (TUG-650)	Data (SpaceWire)	IEEE 1355
10	X-band → Patch TX antenna	RF coaxial	CCSDS 401.0-B

Interface	Description	Standard
ASL-H / ASL-STD	HEO coupling for translunar transport	IDSS (NASA-STD-7001)
Cartridges	Transport of 3-4 cartridges per trip	IDSS + structural support
VADS	Transport and deployment in lunar orbit	IDSS + MIL-STD-1553B
MrN	Transport of nuclear microreactor	IDSS + NASA-STD-3001
VLBI surface	Reception of scientific data (NADIR RX)	CCSDS 734.2-B-1 (DTN)
Earth (DSOC)	Primary data downlink	CCSDS 734.2-B-1
Earth (X-band)	Communications backup	CCSDS 401.0-B
TUG-2	Relay coordination in polar orbit	RF X/Ka + DTN protocol

Step	Action	Control
1	TUG-BIG completes 5th transport trip (Trip 5)	Autonomous GNC
2	TUG-BIG remains in lunar polar orbit ($\geq$800 km)	3-axis stabilization
3	Solar panels orient for relay mode (comms power)	Solar tracking
4	NADIR RX (TUG-640) activates and searches for surface optical beacon	Autonomous acquisition
5	DTN handshake with VLBI surface. Custody transfer established	CCSDS 734.2-B-1
6	During pass ($\sim$30 min): RX data at 50–100 Gbps $\rightarrow$ SSD	Internal buffer
7	Outside pass ($\sim$65 min): DSOC points to Earth and downloads	Orbital planning
8	TUG-2 relieves in opposite orbit. Coverage handoff	RF coordination

Note: Two TUG-BIG in polar orbit, separated 180°, guarantee continuous coverage. Each TUG covers $\sim$30 min of pass over the VLBI site and downloads to Earth during the remaining $\sim$65 min of the orbit. The TUG-2 (DWG-TUG2-00) can be integrated as a third relay node for redundancy.

Mechanism	Implementation
TMR (Triple Modular Redundancy)	Avionics: 3 logic replicas with majority voting. Mitigates Single Event Upsets.
EDAC (Error Detection and Correction)	SSD: Reed-Solomon code (255,223). Corrects up to 16 errors per block.
Autonomous watchdog	External DSP monitors avionics. Automatic restart in less than 100 ms upon latch-up.
SSD shielding	Tantalum 5 mm + Al 6061 10 mm. Reduces total ionizing dose by 90%.
Downlink redundancy	Primary optical DSOC + secondary X-band. Automatic switching upon link loss.
DTN Custody Transfer	The TUG confirms reception of each bundle. Without ack, surface retains and retransmits.
Dual buffer	Surface SSD (32 TB) + TUG SSD (48 TB). Combined autonomy $\sim$6.9 h at 25.6 Gbps.
Redundant engines	8 NEXT-C engines. Normal operation with 6. Up to 2 engines can fail without mission loss.
Solar panels	2 independent panels with bypass diodes. One panel can power critical systems.

ID	Standard	Application
NASA-STD-6016	Materials and processes for structures	Chassis, tanks, panels
NASA-STD-3001	Propulsion and energy systems	Xenon, battery
NASA-STD-5017	Deployment and docking mechanisms	NEXT-C, IDSS, panels, nadir gimbal
NASA-STD-7001	IDSS docking system	IDSS (androgynous)
NASA-STD-5003	Fracture, control and shielding	Panels, tanks, Al/Ta
NASA-STD-8739.8	Software assurance	Avionics, SSD, GNC, DTN router
NASA-STD-5001	Safety factors	SF 2.0 yield / 1.4 ultimate
IEEE 1355	SpaceWire	Data bus
AIAA S-114	Cryogenic couplings	FQD
CCSDS 734.2-B-1	DTN Bundle Protocol	Communications (nadir RX + DSOC)
CCSDS 401.0-B	RF and Modulation	X-band + S-band TT&C
MIL-STD-1553B	Critical data bus	Telemetry, engine control

- **DRY MASS:** 1,610.5 kg (+15.0 kg vs Rev. C due to addition of TUG-640, TUG-650 and SSD upgrade).
- **XENON:** 300 kg per trip. Global margin: +405.7 kg (27%) over 5 trips. Ample.
- **BATTERY:** 45 kg (12 kWh) for eclipse buffer and engine start.
- **SSD (UPGRADED):** 1.0 kg (48 TB, rad-hard, RAID 1). Al/Ta shielding of 5 mm (2.5 kg). Autonomy: 2.6 orbits at 25.6 Gbps, 3.3 orbits at 20 Gbps, 6.7 orbits at 10 Gbps.
- **DSOC (TUG-630):** 5.0 kg. Primary link to Earth (10 Gbps, 1.55 μm). Points to Earth with precision gimbal.
- **NADIR RX (TUG-640, NEW):** 10.0 kg. Nadir optical terminal to receive data from the VLBI surface during passes of $\sim$30 min. Aperture 300 mm, FOV $\pm$60°, 2-axis gimbal, 50–100 Gbps. DTN protocol with *custody transfer*.
- **RF X-BAND (TUG-650, NEW):** 4.0 kg. For downlink to Earth if DSOC fails. 500 Mbps in X-band + TT&C in S-band (10 Mbps). Low-gain patch antenna (does not require fine pointing).
- **ENGINES:** 8x NEXT-C, Isp = 4,190 s, unit thrust 80 mN. Nominal operation with 6 engines (2 in reserve).
- **POWER:** Solar panels of 50 kW total (8 engines $\times$ 7.0 kW + margin). Comms consumption: $\sim$50 W (DSOC + NADIR + X-band + avionics).
- **TRAJECTORY:** Low-Energy Transfer (CJ = 3.170) via L1. Transit time: $\sim$14 days outbound, $\sim$10 days return. In relay mode: lunar polar orbit $\geq$800 km.
- **REUSABILITY:** 100% of the fleet returns to HEO (5x TUG-BIG + 3x TUG-2). In relay mode, the TUGs remain in lunar orbit indefinitely as communications nodes.
- **MAXIMUM SAFE CARGO:** 11,500 kg (xenon margin of 5 kg per trip). Rev. D impact: negligible (Δm = 15 kg).
- **RADIATION PROTECTION:** SSD in shielded Al/Ta box (5 mm) for $>$100 krad. Avionics in 10 mm aluminum vault.
- **RELAY WITH TUG-2:** Two TUG-BIG in polar orbit separated 180° + one TUG-2 as backup. Continuous coverage of the VLBI site in PSR.
- **STANDARDS:** NASA-STD-6016, NASA-STD-3001, NASA-STD-5017, NASA-STD-5001, IEEE 1355, CCSDS 734.2-B-1, CCSDS 401.0-B, MIL-STD-1553B.

Engineering drawing - DWG-TUGBIG-00 Rev. D. Complete VLBI RELAY node: SSD 48 TB + DSOC 10 Gbps (Earth) + NADIR RX 50-100 Gbps (surface) + X-band 500 Mbps. Architecture VLBI $\rightarrow$ TUG relay $\rightarrow$ Earth. Validated by simulation TUG-BIG-v1. Ready for CDR.